

"COUNT MEN SKETCH"

Prepare m counters ($m \sim k \log n$)
 $C[1], C[2], \dots, C[m]$
and a hash function H
Use group testing to design H .

Now, when you see $i \in [-n]$
 $C[H(i,1)] += 1$
 $C[H(i,2)] += 1$
 $\vdots$
 $C[H(i, \log n)] += 1$

To report counting results
number i appears ? times
? $\approx \min(C[H(i,1)], \dots, C[H(i, \log n)])$

Group testing
 $x \in \{0,1\}^n$ unknown but contains k ones
 $A = \{a_i\}_{i=1}^m$ testing matrix
 $y \in \{0,1\}^m$ test results
 $y_i = \sum_{j=1}^n A_{ij} x_j$
Use y to infer x

$m \sim k^2 \log n$ if no error allowed
 $m \sim k \log n$ if allow error (introduces info lower bound)

Goal: $m \sim k \log n$ tests
decoding complexity $k \log n$
very few errors

"Popular Approach"
For index $i = 10110$
Take a random number $3 \in \{1,k\}$
 $\Rightarrow$ The i th row of A is
[0000 0000 0010 0000 0000 0...]
 $\hookrightarrow$ the third block
For another index $i = 11011$
Take random number $2 \in \{1,k\}$
 $A_i = [0000 1101 0000 0000 0000 \dots]$

P.S. Better Idea:
Each index chooses TWO random numbers, e.g. $2, 4 \in \{1,k\}$
 $\Rightarrow A_i = [0000 1101 0000 0000 \dots]$

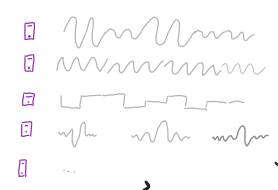
Ideally
Use memory $\sim k \log n$ words
Update time $\sim \log n$
Summary time $k \log n$

Heavy Hitter:
You are given a stream of numbers
3...1...4...1...5...9...2...6...
Tall me k most popular numbers (without remembering all numbers)

Group Testing, Heavy Hitter, & Unsourced Multiple Access
How are they related?
How are they different?

Hsin-Pa Wang
[Folklore]
but check out
[Cheraghchi-Nokas 2006.08.20]

Unsourced Multiple Access Channels



Out of n callphones
 k of them want to talk
using m "frequencies"

Out of n possible messages
 k of them will be sent by $\square$
using m "frequencies"

P.S. No Pilot

$m \sim k \log n$?

decoding complexity $k \log n$?

quantitative GT
semi-quantitative GT
qualitative GT
threshold GT
generalized GT
coin-weighting

Table 1: Many real-world problems are related to Group Testing.

Problem	There are n	k of them	To solve, use m
Disease control [AE22]	people	are sick	virus tests
Genotyping [ECB ⁺ 10]	genes	cause cancer	gene tests
Cognitive radio [LFP22]	cellphones	want to talk	frequencies
Heavy hitter [CM05] / DoS [KGMM08]	users	are spamming	virtual servers
Traitor tracing [CFNP00]	users	resell keys	keys
Computer forensics [GAT05]	files	will be modified	bits of storage
Property-preserving hash [Min22]	properties	appears in a file	bits per file
Image compression [HL00]	wavelet coefficients	are nonzero	bits per digit

Differences

- $x \in \{0,1\}^n$ vs $\mathbb{R}^n$

- $A \in \{0,1\}^{m \times n}$ vs $\mathbb{R}^{m \times n}$

- $y = Ax$ vs "logical matrix multiplication"
- noise model

Statement:
If " $y = Ax$ " is linear / not idempotent
 $\Rightarrow$ More like Compressed Sensing
Can do online peeling / subtraction

If " $y = Ax$ " is idempotent / noisy / forgetful
 $\Rightarrow$ More like Group Testing